

API documentation

6.1 API Documentation

The Nfs platform utilizes **Swagger (OpenAPI)** embedded within the ASP.NET Core 9 framework to automatically generate interactive, real-time API documentation. This ensures seamless integration and reliable communication between the frontend React library and the backend RESTful services.

Core Endpoints Summary:

The system exposes the following primary endpoints to support the platform's functional requirements:

1. **Authentication & User Management** (`/api/auth`)

- `POST /api/auth/register`: Handles secure user registration and dashboard provisioning.
- `POST /api/auth/login`: Validates user/therapist credentials and returns a secure JWT Token.

2. **Therapy Sessions Management** (`/api/bookings`)

- `GET /api/therapists`: Allows patients to browse and filter available mental health professionals.
- `POST /api/bookings`: Enables users to reserve individual 1-on-1 sessions (Online/Offline).

3. **AI Chatbot & Recommendations** (`/api/assessment`)

- `POST /api/assessment/submit`: Processes user responses, invokes the Python AI engine to store anonymized logs in MongoDB, and generates personalized recommendations.

Controller	HTTP Method	Endpoint Route	Description / Use Case
Auth	POST	/api/auth/register	User/Client registration (UC1)
Auth	POST	/api/auth/login	Secure login & JWT Token generation (US2)
Auth	POST	/api/auth/logout	Invalidate token and clear session
Therapists	GET	/api/therapists	Browse and filter therapists (UC2)
Therapists	GET	/api/therapists/{id}/schedule	View available time slots for a therapist
Bookings	POST	/api/bookings	Book a 1-on-1 therapy session (UC2)
Sessions	GET	/api/sessions/group	View available group support sessions (UC3)
Sessions	POST	/api/sessions/group/join	Join group session with nickname/avatar (US6)
Dashboard	GET	/api/therapists/dashboard	View therapist's specific schedule (US10)
AI Engine	POST	/api/ai/chatbot	Interact with AI chatbot & save logs (US7)
AI Engine	POST	/api/ai/recommend	Get personalized

			session suggestions (US8)
Admin	DELETE	/api/admin/management	Admin control over users and sessions (UC6)

screens when code finishes

6.2 Testing & Validation

To ensure the reliability, security, and performance of the Nfs platform, a comprehensive testing strategy was executed. The testing phase focused on validating both functional use cases and the predefined Key Performance Indicators (KPIs).

positive test

Test ID	Component / Feature	Test Scenario / Input	Expected Result	Actual Result	Status	Verified KPI
TC-01	Authentication	User registers with a valid email and strong password.	Account created successfully ; secure JWT token generated.	As Expected	Pass	Security & Reliability
TC-02	Authentication	User attempts to login with incorrect credentials.	System denies access and displays an unauthorized error message.	As Expected	Pass	Security & Reliability
TC-03	System Performance	Execute 50 concurrent simulated API requests (Local Load Test).	API handles requests seamlessly; average response time remains < 500ms.	As Expected (Avg: 320ms)	Pass	Avg API Response < 500ms
TC-04	UI/UX & Frontend	Navigate from Home Page to Therapist Booking Dashboard.	Page completely renders with smooth transitions in less than 3 seconds.	As Expected (Avg: 1.8s)	Pass	Page Load Time < 2-3s
TC-05	UI/UX & Frontend	Client books an individual therapy session.	Completion of the core booking workflow in 3 steps or less.	As Expected (3 Steps)	Pass	Key tasks ≤ 3 steps

TC-06	Session Booking	Client attempts to book an already reserved therapist time slot.	System detects conflict, prevents double-booking, and requests another time.	As Expected	Pass	Successful Request Rate > 95%
TC-07	Privacy & Security	User joins a group support session utilizing a pseudonym/ nickname.	Group chat displays only the nickname and avatar; real identity remains hidden.	As Expected	Pass	Data Privacy & Protection
TC-08	AI Chatbot	Send standard mental health guidance queries to the AI Chatbot.	Chatbot returns relevant, safe responses with ~80% correctness based on predefined scripts.	As Expected (~82% accuracy)	Pass	AI Effectiveness (~80%)
TC-09	AI & Data Logs	Execute AI interaction and inspect the underlying data layers.	AI logs are successfully written to MongoDB with all sensitive user data fully anonymiz			

Negative test

Test ID	Component / Feature	Test Scenario / Input	Expected Result	Actual Result	Status	Verified KPI
TC-10	Validation (Error)	User registers with an email that already exists in the system.	System rejects registration and displays: "Email already exists" (UC1 Alternative Flow).	As Expected	Pass	Successful Request Rate > 95%
TC-11	Booking (Error)	Network failure occurs while the user is confirming a session booking.	System does not freeze; it shows a retry message and keeps data intact.	As Expected	Pass	System Reliability & UX
TC-12	AI Chatbot (Error)	User enters an unsafe or harmful prompt to the chatbot.	Chatbot triggers a safety disclaimer and suggests professional help (UC4 Alternative Flow).	As Expected	Pass	AI Safety & Effectiveness
TC-13	Security (Error)	Unauthenticated user tries to access /api/therapists/dashboard directly.	System blocks the request with a 401 Unauthorized HTTP status code.	As Expected	Pass	Security & Reliability

Deployment Strategy

Write when all agree

